

SOCIOL 723: Practice Midterm, With Answers

For practice before the in-class midterm on Thursday, October 15

Instructions. This practice exam has the same format, length, and level as the midterm. The midterm is open book and open note, timed (150 minutes), and asks for no code. It covers Weeks 1–7 and draws on the frames marked with a blue star in the slides. Where a question asks for sentences, write them as you would in a paper. Show your arithmetic: a right number with no work earns partial credit only. The gray tag after each question names the slides to review (Week 1 has two decks; “Tue” is Tuesday’s).

Part	Format	Weeks	Points
1	Five yes-or-no questions	1–7	10
2	Ten single-choice questions	1–7	30
3	Three longer questions	1–2, 3–4, 5 and 7	60
Bonus	One derivation	7	+10

Part 1: Yes or No

Circle YES or NO. Two points each.

1. Suppose the true model is $\log(\text{earnings}_i) = \beta_0 + \beta_1 \text{schooling}_i + \epsilon_i$, where ϵ_i is the true, unobserved error (everything else that affects earnings, such as ability). You estimate it by OLS and find that the fitted residuals $\hat{\epsilon}_i$ are uncorrelated with schooling. This is evidence that $E[\epsilon_i \mid \text{schooling}_i] = 0$. [Week 1 Tue, slide 15–16] YES NO

Answer. No. The normal equations force $\sum_i \text{schooling}_i \hat{\epsilon}_i = 0$ (and $\sum_i \hat{\epsilon}_i = 0$) in every sample, whatever the true relation between schooling and ϵ_i . The zero correlation is algebra, not evidence. If ability raises earnings and goes with schooling, ϵ_i is correlated with schooling, yet the residuals $\hat{\epsilon}_i$ are still uncorrelated with it: OLS has absorbed that correlation into a biased $\hat{\beta}_1$. The assumption is about the unobserved ϵ_i , and the data cannot check it.

2. In the same regression, the spread of log earnings is wider among the more schooled, so the errors are heteroskedastic. This biases the OLS schooling coefficient. [Week 2, slide 19] YES NO

Answer. NO. Unbiasedness needs only $E[\epsilon_i | X_i] = 0$; it says nothing about the variance of ϵ_i . Heteroskedasticity makes the classical standard-error formula wrong, not the coefficient. Keep $\hat{\beta}$ and report a robust (sandwich) standard error.

3. You fit a logit of voting on age, then fit it again with education added. The log-likelihood at the MLE of the second model can never be lower than that of the first, so the second model will always have the lower (better) AIC. [Week 3, slide 55, 69] YES NO

Answer. NO. The first half is true: the first model is the second with the education coefficient fixed at zero, so the larger model's maximum is at least as high and $LR = 2(\ell_{\text{big}} - \ell_{\text{small}}) \geq 0$. But $AIC = -2\ell + 2k$: adding education lowers -2ℓ by LR and raises the penalty by 2. AIC falls only if $LR > 2$. Example: $\ell = -612.4$ with $k = 2$ and $\ell = -611.9$ with $k = 3$ give AIC 1228.8 and 1229.8; the better-fitting model has the higher AIC.

4. Places in a job-training program are assigned by lottery, and every winner enrolls. The difference in mean earnings between winners and losers two years later estimates the ATE. [Week 5, slide 13] YES NO

Answer. YES. This is classic Random Control Trial (RCT) with completely randomized treatment. The lottery makes $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i$, so the baseline selection term $E[Y_i(0) | D_i = 1] - E[Y_i(0) | D_i = 0]$ is zero, $ATT = ATU = ATE$, and the difference in means is unbiased for all three.

5. A scholarship awarded by lottery is used as an instrument for college attendance in a regression of earnings at age 30. Because the lottery is random, the scholarship is a valid instrument. [Week 7, slide 6–7] YES NO

Answer. NO. The lottery buys independence: winning is unrelated to ability or family background. It does not buy exclusion. The money itself can change earnings through channels other than attendance: winners graduate with less debt, can take unpaid internships or lower-paid first jobs, and may enter a different field. The scholarship affects Y other than through D , so the IV estimate mixes the effect of attending with the effect of the cash. Randomization makes an instrument independent, not excludable.

Part 2: Single Choice

Circle one letter. Three points each.

1. A researcher regresses children's test scores on neighborhood poverty and omits parental income. Parental income raises test scores, and it is lower in poorer neighborhoods. The short regression's poverty coefficient is [Week 1 Tue, slide 35]

- (A) biased downward: poverty looks more harmful than it is.
- (B) biased upward: poverty looks less harmful than it is.
- (C) unbiased, because parental income only affects the outcome.
- (D) impossible to sign without the data.

Answer. (A). Bias = $\beta_2 \delta_1$ with $\beta_2 > 0$ (income raises scores) and $\delta_1 < 0$ (income is lower where poverty is higher): the bias is negative.

2. Students sit in 40 schools, and the treatment was assigned by school. You run a bootstrap with $B = 1,000$ draws, each draw resampling individual students with equal probability, and get a standard error of 0.8. The right bootstrap [Week 2, slide 47]

- (A) resamples whole schools with replacement, each bringing all its students.
- (B) resamples students, but with $B = 5,000$ instead of 1,000.
- (C) resamples students within each school.
- (D) is not needed: resampling students is already heteroskedasticity-robust.

Answer. (A). Resample the units the assignment lottery ran over. Resampling students treats them as independent and copies the too-small standard error of the naive formula; more draws only estimate that wrong number more precisely.

3. Five independent Poisson counts: 2, 5, 1, 4, 3. The MLE $\hat{\theta}$ is [Week 3, slide 29]

- (A) 3
- (B) 2.5
- (C) 15
- (D) 0.6

Answer. (A). $\ell(\theta) = \sum_i (y_i \log \theta - \theta - \log y_i!)$, so the score is $\sum_i y_i / \theta - n = 0$, giving $\hat{\theta} = \bar{y} = 15/5 = 3$.

4. In a logit of voting (0/1), the coefficient on holding a college degree is 0.693, so $e^{0.693} = 2$. Which sentence is correct? [Week 4, slide 14]

- (A) Graduates are twice as likely to vote as non-graduates.
- (B) A degree raises the probability of voting by 0.693.
- (C) A degree doubles the odds of voting, other covariates held fixed.
- (D) A degree raises the probability of voting by 50 percentage points.

Answer. (C). e^β multiplies the odds. It is not a ratio of probabilities, and the change in probability depends on where a person starts on the S-curve.

5. You regress log earnings on schooling and age, and the schooling coefficient is 0.10. Then you (i) regress log earnings on age and save the residuals, (ii) regress schooling on age and save the residuals, and (iii) regress the residuals from (i) on the residuals from (ii). The slope in (iii) is [Week 1 Tue, slide 26, 29]

- (A) exactly 0.10.
- (B) close to 0.10, but not equal, because residuals are estimated.
- (C) the slope of log earnings on schooling without age.
- (D) the coefficient on age.

Answer. (A). This is the Frisch–Waugh–Lovell theorem: the coefficient on schooling in the regression with age equals the slope from regressing the part of earnings not explained by age on the part of schooling not explained by age. It holds exactly, in every sample.

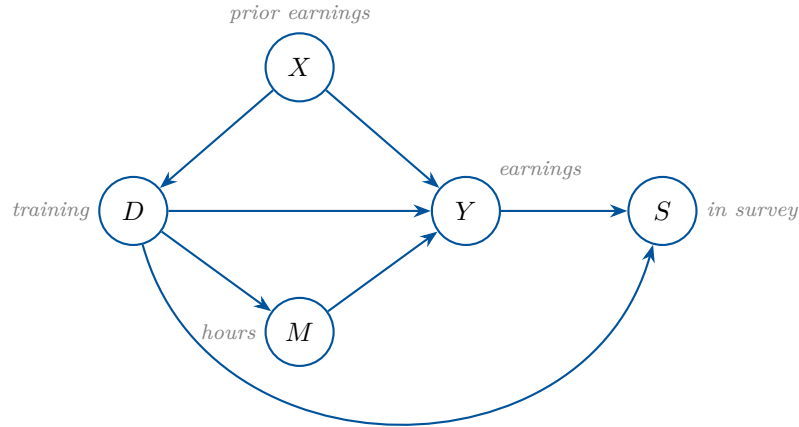
6. Four units: [Week 5, slide 9–10]

Unit	D_i	$Y_i(0)$	$Y_i(1)$
1	1	3	9
2	1	5	9
3	0	4	6
4	0	2	2

- (A) ATE = 3, ATT = 5, naive comparison = 6
- (B) ATE = 5, ATT = 3, naive comparison = 6
- (C) ATE = 3, ATT = 5, naive comparison = 5
- (D) ATE = 4, ATT = 5, naive comparison = 6

Answer. (A). $\tau_i = 6, 4, 2, 0$. ATE = $12/4 = 3$; ATT = $(6 + 4)/2 = 5$. Observed $Y_i = 9, 9, 4, 2$, so naive = $9 - 3 = 6 = \text{ATT} + \text{selection bias} = 5 + (4 - 3)$.

7. In the DAG below, D is job training, Y is earnings two years later, X is earnings before training, M is hours worked after training, and S is whether the person is still in the survey at follow-up. To estimate the *total* effect of D on Y , condition on [Week 5, slide 32, 41, 46]



- (A) X only.
- (B) X and M .
- (C) X and S (that is, analyze only people still in the survey).
- (D) X , M , and S .

Answer. (A). X is a confounder (the back door $D \leftarrow X \rightarrow Y$). M is a mediator: conditioning removes part of the effect. S is a collider ($D \rightarrow S \leftarrow Y$): conditioning, including by keeping only respondents still in the survey, opens a spurious path.

8. In a cell where the propensity score is $e(x) = 0.8$, weighting for the ATT gives each treated unit the weight 1 (the treated are already the population we want). Each *untreated* unit gets the weight [Week 6, slide 41]

- (A) 4
- (B) 1.25
- (C) 5
- (D) 0.25

Answer. (A). ATT weights: treated 1, untreated $e(x)/(1 - e(x)) = 0.8/0.2 = 4$, the odds of treatment. The untreated are reweighted to look like the treated, of whom there are four per untreated unit in this cell.

9. Two cells of a covariate, with ignorability given the cell: [Week 6, slide 8]

cell	treated (n)	untreated (n)	treated mean Y	untreated mean Y
1	10	30	8	5
2	30	30	12	10

The ATE from exact matching is

- (A) 2.4
- (B) 2.25
- (C) 2.5
- (D) 3.5

Answer. (A). Within-cell differences $8 - 5 = 3$ and $12 - 10 = 2$. The ATE weights cells by their share of *everyone*: $(40 \times 3 + 60 \times 2)/100 = 2.4$. The ATT weights by the treated, $(10 \times 3 + 30 \times 2)/40 = 2.25$; the unweighted average of the two cells is 2.5; the naive comparison is $11 - 7.5 = 3.5$, inflated because the treated sit mostly in cell 2.

10. A regression of wages on schooling uses one instrument whose first-stage robust F is 2.5 in a sample of 3,000 men. Compared with OLS, the 2SLS estimate [Week 7, slide 21–22, 65]
- (A) is pulled toward the OLS estimate, and its usual 95% interval covers the truth less than 95% of the time.
 - (B) is unbiased; the only cost of a weak instrument is a wider interval.
 - (C) is pulled toward zero, as under classical measurement error.
 - (D) is fine, because the sample is large and the reduced form is what matters.

Answer. (A). With a weak instrument the first stage fits sampling noise that is correlated with the structural error, so 2SLS is centered between the truth and OLS, and its sampling distribution is far from normal: the usual interval is too narrow. Report the Anderson–Rubin interval, which keeps its size at any F , together with the first stage and the reduced form.

Part 3: Longer Questions

Twenty points each. Every number below comes from the course's GSS extract or from Lab 7's data.

Question 1: Schooling, Parental Education, and Earnings

The GSS extract has 3,509 full-time workers aged 25–64 (GSS 2010–22). The outcome is log annual earnings. Three OLS regressions:

	(1) log earnings	(2) log earnings	(3) parents' education
years of schooling	0.1187	0.1099	0.5703
parents' education (years)		0.0154	
intercept	8.2307	8.1546	4.9405
R^2	0.129	0.132	

Standard errors for the schooling coefficient in model (2), three ways: classical 0.0058; HC3 0.0064; clustered on the GSS primary sampling unit (839 clusters, median size four) 0.0065.

- (a) Interpret the schooling coefficient in model (1) in one sentence a reader of a sociology journal would understand. [Week 1 Tue, slide 23]

Answer. Among these workers, each additional year of schooling is associated with about 0.119 log points, roughly 12%, higher annual earnings. It is a description (the best linear approximation to the CEF), not yet a causal claim.

- (b) Use the omitted-variable-bias formula to recover the coefficient in model (1) from models (2) and (3). Show the arithmetic, and explain in words why adding parents' education lowers the schooling coefficient. [Week 1 Tue, slide 32–35]

Answer. Short = long + (effect of omitted) $\times$ (regression of omitted on included): $0.1099 + 0.0154 \times 0.5703 = 0.1099 + 0.0088 = 0.1187$. Parents' education raises earnings ($0.0154 > 0$) and moves with own schooling ($0.5703 > 0$), so model (1) credits schooling with part of the family-background advantage; the bias is positive.

- (c) Is 0.1099 the causal effect of a year of schooling? Name one variable still omitted, and explain the bias it creates. [Week 1 Tue, slide 39, 67–68]

Answer. No, not without arguing that nothing else confounds schooling and earnings. Example: ability. It raises earnings and goes with more schooling, so by the OVB formula (positive $\times$ positive) the coefficient is still biased upward.

- (d) Using the HC3 standard error, compute the t -statistic and a 95% confidence interval for the schooling coefficient in model (2). [Week 2, slide 17, 30]

Answer. $t = 0.1099/0.0064 = 17.2$. Interval: $0.1099 \pm 1.96 \times 0.0064 = 0.1099 \pm 0.0125 = [0.097, 0.122]$.

- (e) The GSS samples areas first and then people within them. Which of the three standard errors would you report? [Week 2, slide 41]

Answer. The clustered one: cluster at the level of the sampling lottery, the primary sampling unit. Here the clusters are tiny (median four people), so little within-cluster correlation can accumulate and the clustered SE barely moves (0.0065 vs 0.0064). Say so rather than choosing the smaller number.

- (f) A colleague suggests adding occupational prestige to model (2) “to be safe.” Do you agree? Answer with a reason that refers to what the coefficient would then estimate. [Week 1 Tue, slide 72]

Answer. No. Occupation is a consequence of schooling on the path to earnings, a mediator. Holding it fixed removes the part of the schooling effect that works through better jobs, so the coefficient would no longer be the total effect. More controls are not more conservative.

Question 2: A Logit of Ever Having Married

Same 3,509 workers; 73% have ever married. A logit of ever married (0/1):

	$\hat{\beta}$	SE	z
intercept	-2.23	0.279	-7.98
years of schooling	-0.01	0.015	-0.99
female	0.04	0.084	0.49
Black	-0.96	0.107	-9.04
age (years)	0.09	0.004	19.91
log-likelihood	-1,769.5		

- (a) Write the model twice: once in log-odds form and once in probability form. [Week 4, slide 9]

Answer. $\log \frac{p_i}{1-p_i} = \beta_0 + \beta_1 \text{school}_i + \beta_2 \text{female}_i + \beta_3 \text{Black}_i + \beta_4 \text{age}_i = X_i' \beta$, and $p_i = e^{X_i' \beta} / (1 + e^{X_i' \beta})$.

- (b) Compute, by hand, the predicted probability of having married for a woman with 16 years of schooling who is not Black, at age 30 and at age 40. [Week 4, slide 15–16]

Answer. Age 30: $X' \hat{\beta} = -2.23 - 0.01(16) + 0.04 + 0.09(30) = -2.23 - 0.16 + 0.04 + 2.70 = 0.35$; odds $e^{0.35} = 1.42$; $\hat{p} = 1.42/2.42 = 0.59$. Age 40: add $10 \times 0.09 = 0.90$, so $X' \hat{\beta} = 1.25$, odds $e^{1.25} = 3.49$, $\hat{p} = 3.49/4.49 = 0.78$.

- (c) For this woman, ten years of age raise the probability by about 19 points. The average marginal effect of age is 0.014 per year, so ten times the AME is only 14 points. Explain the difference. [Week 4, slide 17–18]

Answer. The slope of the probability is $\hat{p}_i(1 - \hat{p}_i)\hat{\beta}$: steepest at $\hat{p} = 0.5$. This woman starts at 0.59, near the steep part of the S-curve. The AME averages the slope over all 3,509 people, and with 73% married most of them sit higher, where the curve is flatter.

- (d) Adding eight census-region dummies raises the log-likelihood to -1,756.6. Compute the BIC of both models, $BIC = -2\ell + k \log n$, with $k = 5$ and $k = 13$ parameters and $\log n = 8.16$. Under the BIC criterion, should the region dummies be included? [Week 3, slide 70–71]

Answer. Without region: $-2(-1769.5) + 5 \times 8.16 = 3539.0 + 40.8 = 3579.8$. With region: $-2(-1756.6) + 13 \times 8.16 = 3513.2 + 106.1 = 3619.3$. Smaller is better, so BIC says no: the eight dummies bought 25.8 units of fit, about 3.2 each, less than BIC's price of 8.16 per parameter.

Question 3: College Proximity as an Instrument (Card 1995)

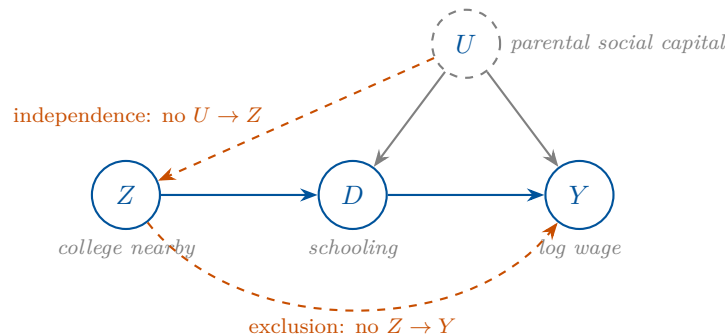
Lab 7's data: 3,010 men from the NLS Young Men, first interviewed in 1966. D = years of schooling; Y = log hourly wage in 1976; $Z = 1$ if the man grew up in a county with a four-year college (68% did). All four regressions control for experience, experience squared, Black, metropolitan residence, and South. HC1 standard errors in parentheses:

	OLS of Y on D	first stage, D on Z	reduced form, Y on Z	IV
estimate	0.074	0.337	0.045	0.132
(se)	(0.004)	(0.081)	(0.016)	(0.049)

Robust first-stage $F = 17.5$. (Unrounded: reduced form 0.0446, first stage 0.3373.)

- (a) Draw a DAG with Z , D , Y , and unmeasured parental social capital U (the parents' networks and connections), which affects both schooling D and the son's wage Y . Include the two arrows whose presence would violate independence and exclusion, and mark them as the ones that must be absent. [Week 7, slide 6–7]

Answer. Solid arrows are the design; the dashed orange arrows must be absent.



Present: $Z \rightarrow D \rightarrow Y$; $U \rightarrow D$ (well-connected parents steer their sons into more schooling); $U \rightarrow Y$ (the same connections help them find better-paid jobs). These two arrows are why OLS is biased and an instrument is needed. Must be absent: $Z \rightarrow Y$, any effect of proximity on wages that does not run through schooling (exclusion); and $U \rightarrow Z$, parental social capital shaping where the family lives (independence).

- (b) For each of the two conditions, give one concrete reason the marked arrow might in fact be present in Card's setting. [Week 7, slide 30]

Answer. Independence ($U \rightarrow Z$): families do not land near colleges at random. Parents with more social capital (professional networks, ties to institutions) are more likely to live in college towns, and the same connections raise their sons' wages; then Z is

correlated with the error. Card’s controls (metropolitan, South) and family background variables only partly close this. **Exclusion** ($Z \rightarrow Y$): counties with a college have different labor markets, more professional jobs and higher wages, which raise a man’s wage whether or not he attended. Neither can be tested; both have to be argued.

- (c) Show that the IV estimate is the ratio of two numbers in the table, and interpret it in one sentence. [Week 7, slide 8–10, 31]

Answer. $\hat{\beta}_{IV} = \text{reduced form}/\text{first stage} = 0.0446/0.3373 = 0.132$. Growing up near a college raises wages by 4.5% and schooling by a third of a year; scaled up, a year of schooling raises wages by about 13% for the men whose schooling proximity changed.

- (d) Omitted parental social capital predicts that OLS overstates the return to schooling, yet IV (0.132) is almost twice OLS (0.074). Give two explanations consistent with valid instruments. [Week 7, slide 16, 28]

Answer. (i) **Measurement error** in reported schooling pulls OLS toward zero; IV is not affected by classical error in D . (ii) **LATE**: IV estimates the return for compliers, men whose schooling was moved by having a college nearby, typically from poorer families facing cost barriers; their return can be higher than the average return OLS summarizes. (A violated exclusion or weak-instrument bias would also push IV up, but those are failures of the design, not explanations under valid instruments.)

- (e) Treat schooling as “attended college or not.” Describe compliers, always-takers, and never-takers in this setting, and state monotonicity in words. Is the IV estimate the right number for a policy that opens new colleges in counties without one? [Week 7, slide 12–13, 16–17]

Answer. Compliers attend college only if one is nearby; always-takers attend wherever they grow up; never-takers do not attend either way. Monotonicity: no one attends *because* there is no college nearby. For opening new colleges the LATE is close to the right number: the men such a policy would move are exactly those whose attendance responds to proximity. It is not the effect for always-takers or for everyone.

Bonus: Deriving the Wald Estimate

Ten points. Binary instrument Z_i , binary treatment D_i . Write $D_i(z)$ for the treatment unit i takes when assigned $Z_i = z$, and $Y_i(d)$ for its potential outcomes, so that $D_i = D_i(Z_i)$ and $Y_i = Y_i(D_i)$. Write $\tau_i = Y_i(1) - Y_i(0)$. The four types are compliers ($D_i(1) = 1, D_i(0) = 0$), always-takers ($1, 1$), never-takers ($0, 0$), and defiers ($D_i(1) = 0, D_i(0) = 1$). Assume (i) independence: $(Y_i(0), Y_i(1), D_i(0), D_i(1)) \perp\!\!\!\perp Z_i$; (ii) exclusion, already built into writing $Y_i(d)$ without z ; (iii) relevance: the first stage is not zero. Do *not* assume away defiers until part (c). [Week 7, slide 12–15]

- (a) The denominator. Show that the first stage is $E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0] = \Pr(\text{complier}) - \Pr(\text{defier})$.
- (b) The numerator. Show that the reduced form is

$$E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0] = \Pr(\text{complier}) E[\tau_i \mid \text{complier}] - \Pr(\text{defier}) E[\tau_i \mid \text{defier}].$$

Say where you use exclusion and where you use independence.

- (c) Now assume monotonicity, $D_i(1) \geq D_i(0)$ for every i (i.e., no defiers). Show that the Wald ratio, numerator over denominator, equals $E[\tau_i \mid \text{complier}]$, the LATE.
- (d) Without monotonicity, suppose 20% are compliers with an average effect of 1 and 5% are defiers with an average effect of 3. Compute the Wald ratio and explain why it is not an average effect for anyone.

Answer. (a) You can show it either by expression below, or by plotting two groups $Z = 1$ and $Z = 0$ with different shares of treated units, as we did in class. Same intuitions in the following questions.

When $Z_i = 1$ the unit takes $D_i(1)$, and $D_i(1) = 1$ exactly for compliers and always-takers:

$$\begin{aligned} E[D_i \mid Z_i = 1] &= E[D_i(1) \mid Z_i = 1] && D_i = D_i(Z_i) \\ &= E[D_i(1)] && \text{independence} \\ &= \Pr(\text{complier}) + \Pr(\text{always-taker}). && D_i(1) \text{ is a 0/1 indicator} \end{aligned}$$

The same lines with $Z_i = 0$ give $E[D_i \mid Z_i = 0] = E[D_i(0)] = \Pr(\text{always-taker}) + \Pr(\text{defier})$. Subtracting, the always-takers cancel: $\Pr(\text{complier}) - \Pr(\text{defier})$.

- (b) Since D_i is 0 or 1, $Y_i = Y_i(D_i) = Y_i(0) + D_i \tau_i$; this is where exclusion enters, because

$Y_i(d)$ carries no z . Then

$$\begin{aligned} E[Y_i \mid Z_i = 1] &= E[Y_i(0) + D_i(1) \tau_i \mid Z_i = 1] && \text{switching equation, } D_i = D_i(1) \\ &= E[Y_i(0) + D_i(1) \tau_i], && \text{independence: the bracket does not involve } Z_i \end{aligned}$$

and likewise $E[Y_i \mid Z_i = 0] = E[Y_i(0) + D_i(0) \tau_i]$. Subtracting, $E[Y_i(0)]$ cancels and the reduced form is $E[(D_i(1) - D_i(0)) \tau_i]$. The difference $D_i(1) - D_i(0)$ is 1 for compliers, -1 for defiers, and 0 for always- and never-takers, so by iterated expectations over the four types

$$E[(D_i(1) - D_i(0)) \tau_i] = \Pr(\text{complier}) E[\tau_i \mid \text{complier}] - \Pr(\text{defier}) E[\tau_i \mid \text{defier}].$$

(c) Monotonicity rules out $D_i(1) = 0, D_i(0) = 1$, so $\Pr(\text{defier}) = 0$. The numerator becomes $\Pr(\text{complier}) E[\tau_i \mid \text{complier}]$ and the denominator $\Pr(\text{complier})$, which is positive by relevance. Their ratio is $E[\tau_i \mid \text{complier}]$: the average effect among the people the instrument moves.

(d) Numerator $0.20 \times 1 - 0.05 \times 3 = 0.05$; denominator $0.20 - 0.05 = 0.15$; Wald ratio $0.05/0.15 = 0.33$. Every person the instrument moves has an effect of at least 1, yet the ratio is 0.33: defiers enter with a negative weight, so the ratio is a difference of two group averages, not an average over any group. With the defiers' effect at 4 or more the ratio would be zero or negative.